

TANISHQ RAWAT

Ujjain, Madhya Pradesh

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Education

Acropolis Institute of Technology and Research, Indore

2019-2023

Bachelor of Technology in Computer Science and Engineering (CGPA of 8.10)

Madhya Pradesh, India

Technical Skills

Languages & DB: Python, JavaScript, C++, Shell, PostgreSQL, Redis, Qdrant

Frameworks: FastAPI, Pandas, Polars, Numpy, celery, jQuery, Django

Platforms & Tools : Azure, AWS, Kafka, Git, Docker, Github Copilot, OpenAI, Langchain

Area of Interest: Backend Development, Generative AI, RAG

Experience

Simform Solutions

Feb 2025 – Present

Software Engineer

Remote

- Architected a multi-tenant **FastAPI backend** using a strict repository + service pattern (SOLID-aligned), reducing code change ripple effects by **40%**.
- Optimized API performance via query path refactoring, pagination, and selective serializer variants, cutting average **P95 response time** from **480 ms** to **210 ms** and reducing DB query count per request by **55%**.
- Built an event-driven notification framework over **Redis Streams**, enabling real-time tenant alerts, audit trails, and reactive data pipelines (**latency ↓ 65%**, **scalability ↑ 3x**).
- Designed and implemented a robust **role-based access control (RBAC)** system with centralized authorization policies, strengthening security and simplifying permission management.
- Integrated **Azure Functions** and **Graph API** utilities for asynchronous tenant operations, improving throughput capacity by **45%** under peak load.
- Developed an LLM-powered document intelligence pipeline to extract structured knowledge from unstructured documents and ingest it into reporting systems, enabling real-time analytics and improving data availability for business insights.
- Developed a scalable Document Manager service to handle thousands of documents, implementing centralized functionality for secure storage, retrieval, versioning, and lifecycle management in Azure Blob Storage, improving maintainability and reducing operational overhead by **54%**.

Mobileum Inc.

Aug 2022 – Feb 2025

Software Engineer (Apr 2024 – Feb 2025)

Remote

- Designed and implemented a scalable document processing pipeline leveraging **LLMs** for context-aware tagging, categorization, and metadata extraction across heterogeneous document formats.
- Designed and implemented an ORM-driven DataModel layer to automate data injection into dashboards, eliminating manual query handling and improving development efficiency.
- Implemented transactional workflows and **idempotent service** methods for complex annuity operations, improving data integrity and eliminating race-condition defects (**error rate ↓ 30%**).

Associate Software Engineer (Aug 2022 - Mar 2024)

Bangalore, India

- Engineered a modular orchestration workflow engine enabling **no-code ETL pipeline design**, empowering teams to build multi-stage data flows with visual drag-and-drop nodes and reducing manual integration effort by **60%**.
- Worked on caching mechanism for DataModel helping in dashboard development resulting in decreased latency.
- Automated various utility tasks related to data dump from different data sources using shell script python

Cerebry

Aug 2021 – Feb 2022

AI Software Developer

Singapore, Remote

- Led the development of a framework to extract text from images and convert it into Latex, enabling automated question generation.
- Developed backend modules to generate parameterized LaTeX representations of math questions, enabling dynamic question creation and consistent formatting.

Projects

Online Judge - Code Execution on Scale | *GenAI, FastAPI, Docker, Celery, Redis* | [Source Code](#)

- Designed and implemented a **secure code-execution service** that runs user-submitted solutions inside **isolated Docker containers** to prevent crashes, infinite loops, or excessive memory usage from affecting the main application.
- Built a scalable evaluation pipeline using **FastAPI, Redis, and Celery** workers to handle real-time problem submissions, enforce resource limits, and reliably return test-case results similar to platforms like **LeetCode/HackerRank**.

Cloud On Tips | *Python, Django, ORM, Celery, AWS, Flask, Boto3, jQuery* | [View Demo](#) | [Source Code](#)

- A web application to create & manage **VPS & Serverless Computing Services** without having knowledge to configure it using cloud services. You can create a custom VPS, start, stop, or terminate it whenever required with a single click.
- AWS EC2 instances are being served as VPS and configured using **boto3 SDK**. This platform automates all the actions for EC2. **Serverless Computing Services are built from scratch with no use of AWS LAMBDA**.

Achievements

- Published python package on pypi for logging [**Visit:** <https://pypi.org/project/loggingontips/>]
- **Star Performer Award** recipient at **Mobileum** for my exemplary contribution in Quarter 1 & 3 for year 2023
- Qualified for **Semi-Finals** of **FlipKart Grid 4.0** in 2022 and **Uber HackTag** in 2022
- Achieved **State Rank 1** (GOLD Medalist) in IOM 2019 in Stage 1 and **State Rank 3** in Stage 2
- Solved over **550 problems** on **Leetcode** (Username: tanishqrawat8)
- Wrote **research paper** on creating Custom Web Server in C++ & Automating ticket system via QR Ticketing System